SYKES-KALI

Dedicated Kali Linux Security Lab Machine

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Project Overview

SYKES-KALI is a repurposed ASUS X550Z laptop converted into a dedicated Kali Linux security lab machine. Originally found in storage running Windows 10 with a failing HDD, the system was diagnosed, hardware-upgraded, and fully rebuilt as a networked penetration testing and cybersecurity practice node accessible via SSH from a remote Windows workstation.

This project demonstrates hands-on competency in hardware repair, Linux installation, BIOS configuration, network setup, and remote system administration, all core skills for IT and cybersecurity roles.

Hardware Specifications

Device	ASUS X550ZA (X550ZA-RB11)
CPU	AMD A10-7400P Radeon R6 (4 Cores, up to 3.2GHz)
RAM	8GB DDR3 SODIMM (~6.9GB usable)
Storage	Fanxiang S101Q 256GB SATA SSD (upgraded)
GPU	AMD Radeon R6 Integrated Graphics
Wi-Fi	MediaTek MT7630E 802.11bgn
Ethernet	Realtek RTL8111 Gigabit
OS	Kali Linux 6.18.12 (UEFI Mode, Xfce Desktop)
Hostname	SYKES-KALI
Username	bsykes

Build Process

Phase 1 — Hardware Assessment

The laptop was pulled from storage and booted to assess viability. Key findings:

- BIOS confirmed AMD A10-7400P, 8192MB RAM, better specs than initially assumed
- CrystalDiskInfo SMART check on the original WD Blue 1TB HDD returned Health Status: GOOD with zero reallocated, pending, or uncorrectable sectors

- 100% disk usage at idle was attributed to Windows 10 behavior on a slow HDD, not drive failure
- Trackpad non-functional (hardware damage from prior virus infection)
- Documents folder found encrypted from a 2019 ransomware infection (Globe/Purge variant) files unrecoverable, drive wiped

Photo 1 — ASUS X550Z closed lid, original condition pulled from storage

Photo 2 — Bottom panel removed, WD Blue 1TB HDD visible alongside motherboard and cooling system

Phase 2 — SSD Swap

The original WD Blue 1TB HDD was replaced with a Fanxiang S101Q 256GB SATA SSD. The stripped retaining screw was addressed before removal. The SSD was mounted in the existing 2.5" SATA drive bay using the original bracket.

- 256GB chosen for adequate space: Kali full install (~20GB) + tools + wordlists (SecLists ~1GB+)
- SATA SSD format compatible with X550Z's SATA-only drive bay
- Original 1TB HDD reformatted (NTFS, full format, 32KB allocation units) and repurposed as a Jellyfin anime backup drive in a USB enclosure

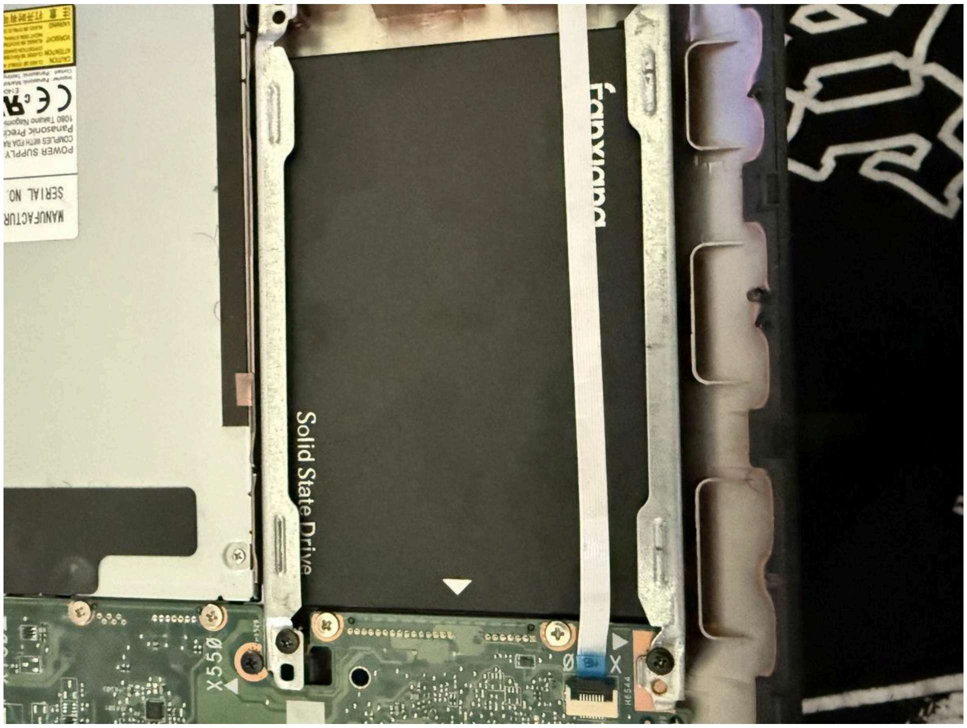

Photo 3 — Fanxiang 256GB SATA SSD installed in the drive bay with mounting bracket

Phase 3 — BIOS Configuration

On first boot with the new SSD, the system entered BIOS setup as expected with no OS present. The following configuration changes were made:

- Confirmed UEFI boot mode (Launch CSM: Disabled)
- Disabled Fast Boot to allow USB installer detection
- Disabled Secure Boot to allow Kali Linux unsigned bootloader
- Verified system date, RAM, and CPU recognized correctly

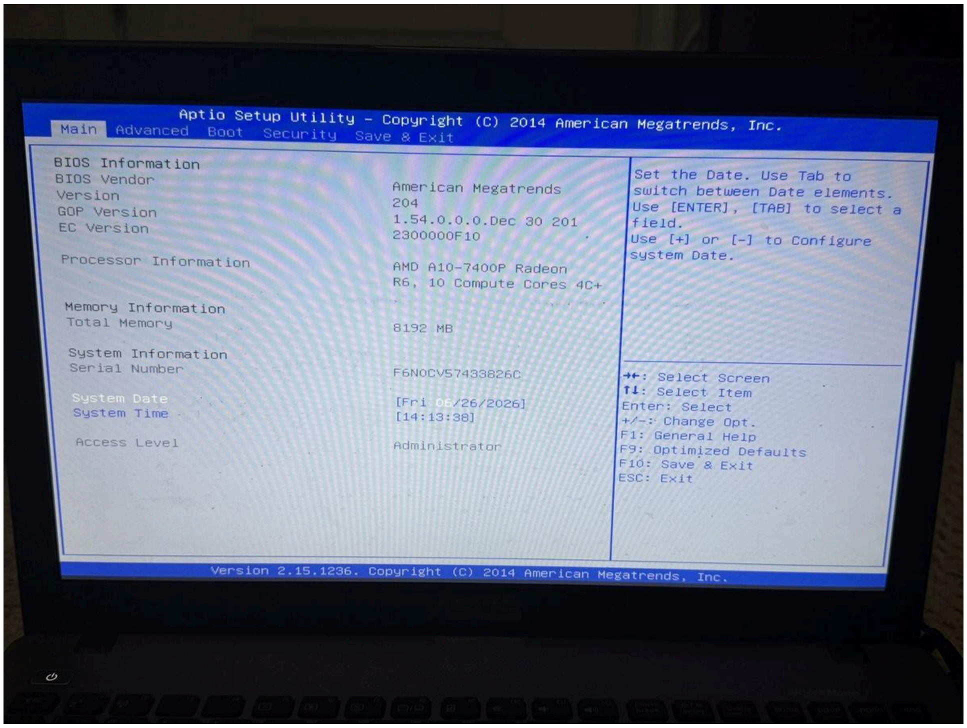

Photo 4 — BIOS POST screen confirming AMD A10-7400P, 8192MB RAM, June 26 2026

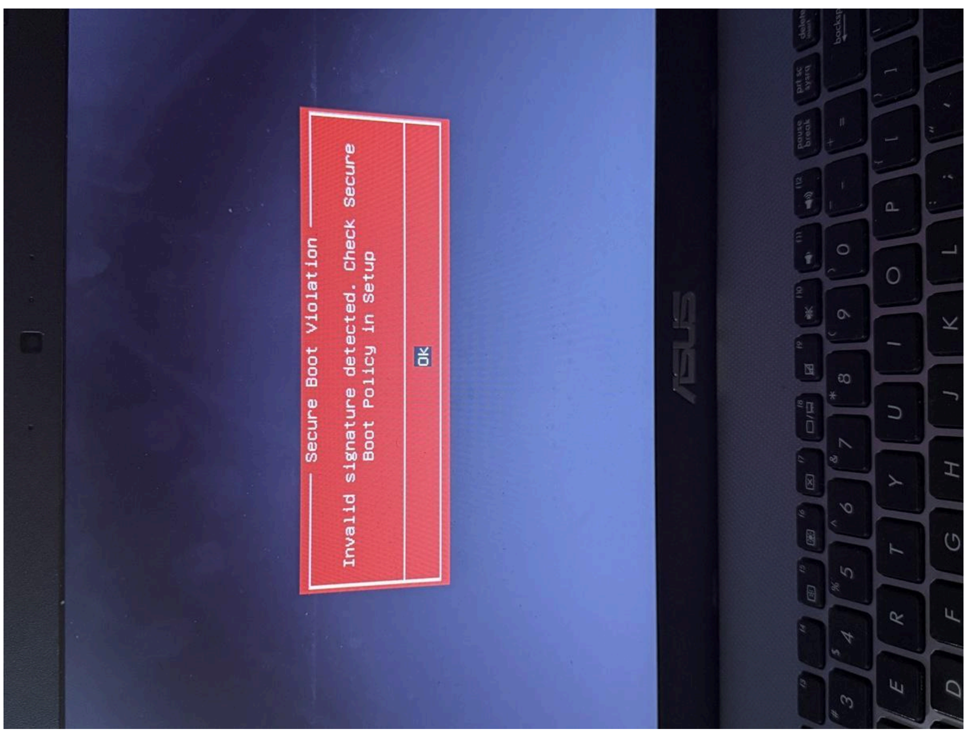

Photo 5 — Secure Boot Violation error on first USB boot attempt. Resolved by disabling Secure Boot in BIOS Security tab

Phase 4 — Kali Linux Installation

Kali Linux was installed from an existing ISO flashed to a USB drive via balenaEtcher (GPT/UEFI mode). The installation was performed using the Graphical Install option.

- Partition scheme: Guided - Entire Disk, All files in one partition
- Filesystem: ext4 (main), ESP (EFI), swap
- Network: Skipped during install (no ethernet cable available) configured post-install via WiFi
- Desktop environment: Xfce (lightweight, optimal for A10-7400P hardware)
- Tool packages: top10 + default recommended tools (Nmap, Metasploit, Wireshark, Burp Suite, etc.)
- Hostname set to SYKES-KALI, username bsykes

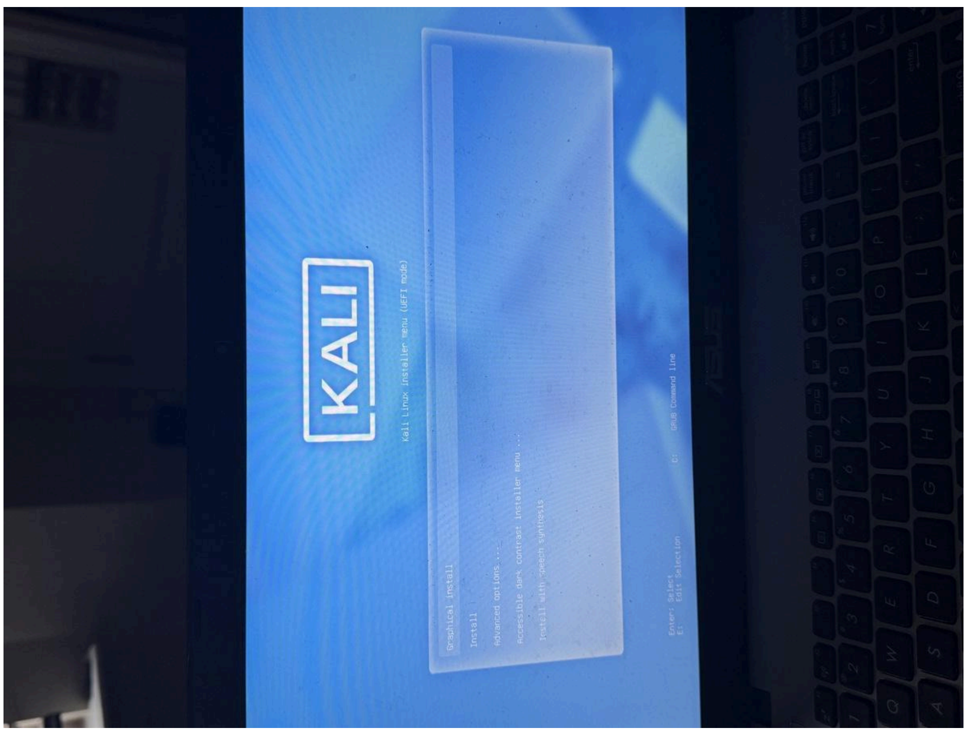

Photo 6 — Kali Linux installer menu in UEFI mode after Secure Boot disabled

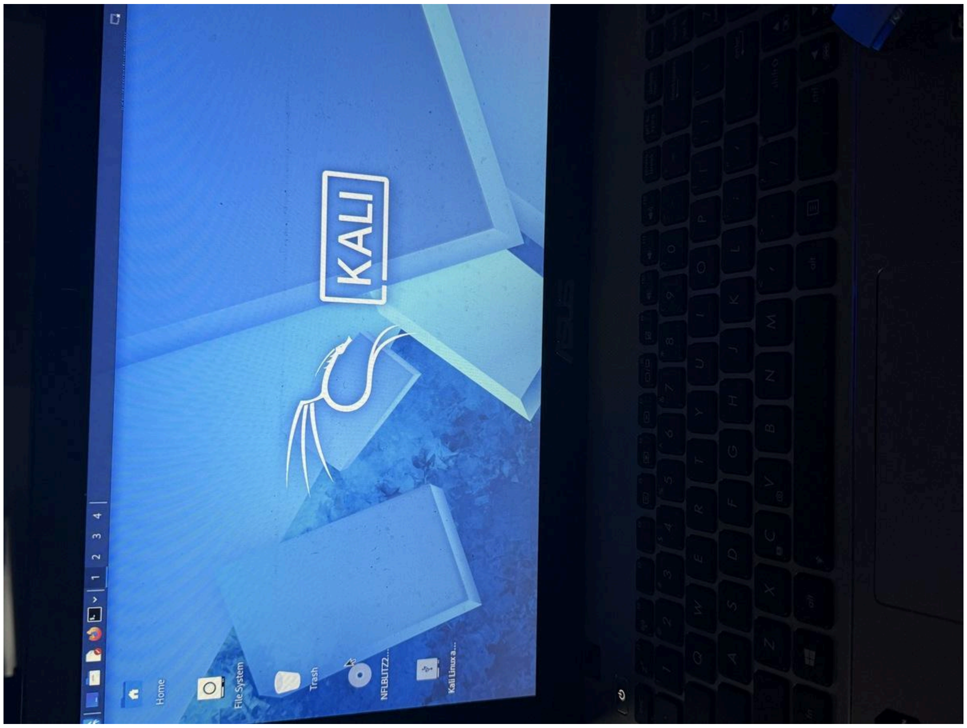

Photo 7 — SYKES-KALI Kali Linux desktop fully booted and operational

Post-Installation Configuration

System Updates

Immediately after boot, full system update was performed:

```
sudo apt update && sudo apt upgrade -y
```

SSH Remote Access

SSH was enabled and set to start on boot, allowing remote management from the main Windows workstation upstairs via PowerShell — eliminating the need to physically interact with the laptop.

```
sudo systemctl enable ssh && sudo systemctl start ssh
# From Windows PowerShell on main PC:
ssh bsykes@192.168.1.224
```

Sleep Disabled

All sleep and suspend targets were masked to prevent dropped SSH sessions:

```
sudo systemctl mask sleep.target suspend.target hibernate.target
hybrid-sleep.target
```

Skills Demonstrated

- Hardware diagnostics — SMART health analysis, component identification, failure triage

- Physical hardware repair — laptop disassembly, drive replacement, stripped screw extraction
- BIOS/UEFI configuration — Secure Boot, CSM, Fast Boot, boot order management
- Linux installation — Kali Linux UEFI install, partitioning, software selection
- Network administration — SSH setup, IP addressing, remote system management
- Security awareness — ransomware identification, malware triage, incident response
- System hardening — sleep/suspend disabled for always-on availability
- Documentation — full build process photographed and documented for portfolio

Planned Use Cases

- Penetration testing practice with Kali tools (Nmap, Metasploit, Burp Suite, Wireshark)
- TryHackMe and HackTheBox lab exercises
- Network scanning and vulnerability assessment against home lab (SYKESHOMESERVER)
- Brute force attack simulations against Active Directory lab environment
- SOC Analyst Tier 1 skill development, log analysis, threat detection
- Post-Security+ certification study and hands-on reinforcement